Seeing is Believing: Exposure to Female National Leaders and Attitudes towards Gender Equality

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Outline

- Background
  - Women Leadership as a Trending Phenomenon
  - Contributors and Consequences
  - Exposure to Women Head of States (WHOS)
- Research Questions
  - Does exposure matters?
  - Leaders – what types of leadership matter?
  - People – who change their minds after exposure?
- Design, Data and Methods
- Findings
- Conclusion and Discussion

Background

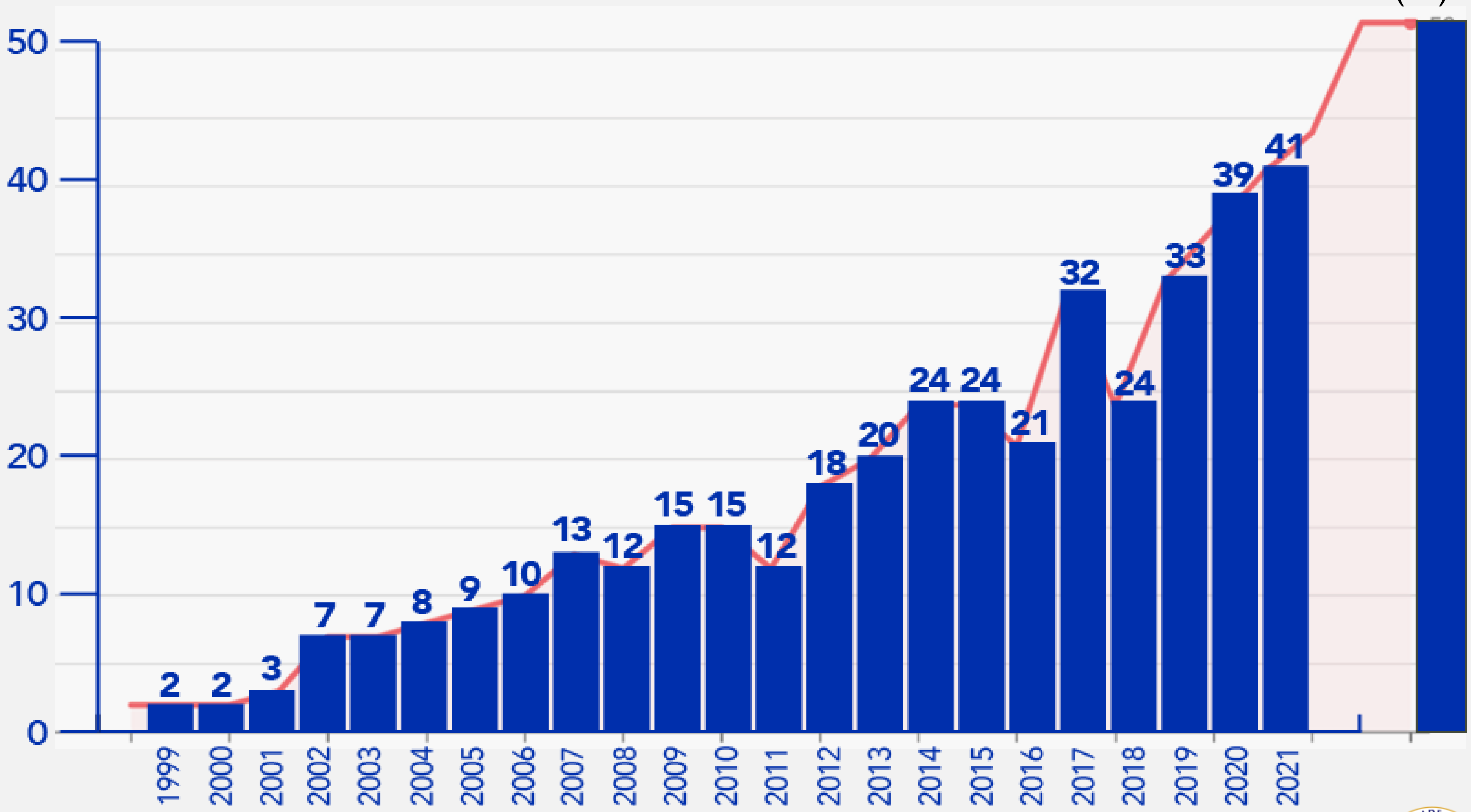
Women Leadership on the Rise

- **Women leadership** refers to the exercise of authority, decision-making, and influence by women in various positions and domains.

- Women leadership has been steadily growing since the mid-20th century in various fields, including but not limited to education, culture and art, science and technologies, business and management, and political sectors.

Number of Women CEOs in Fortune 500 Companies by Year (1999 - September 2020)

Fortune 500 women CEOs as of Sept 2024 (52) 10.4%



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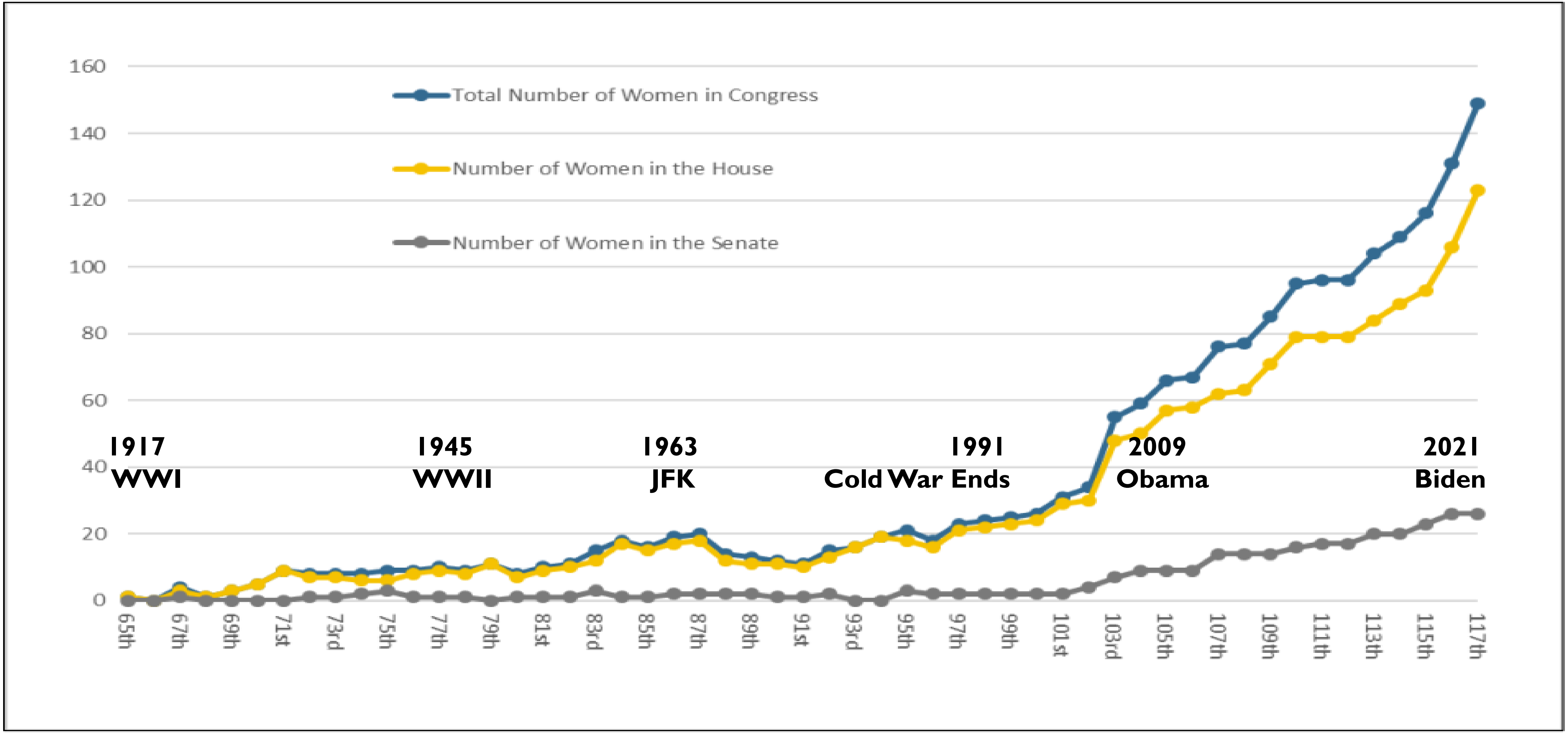
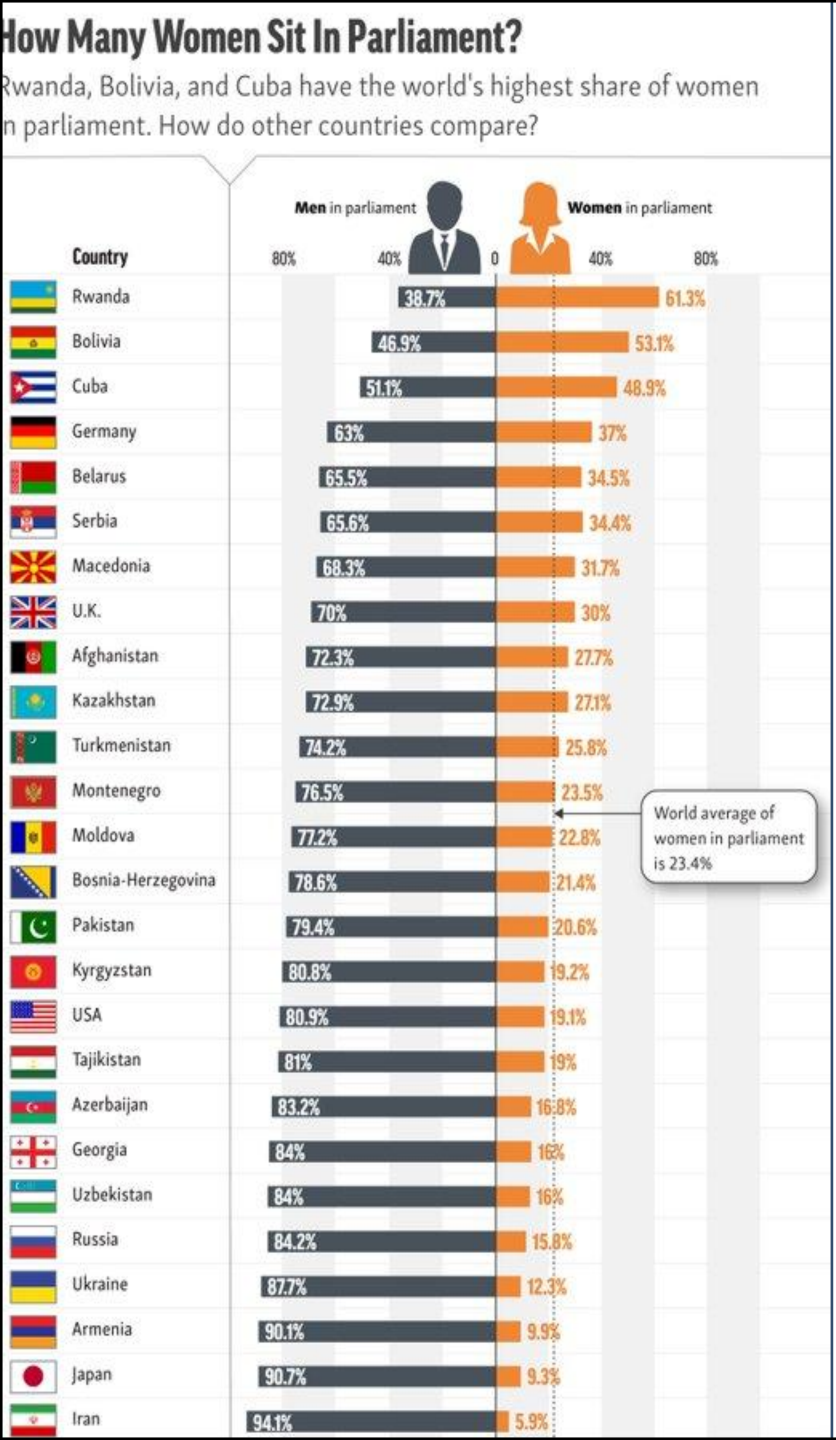
50
Women CEOs
slaying it in 2024

Figure 1. Number of Women by Congress: 1917-2021

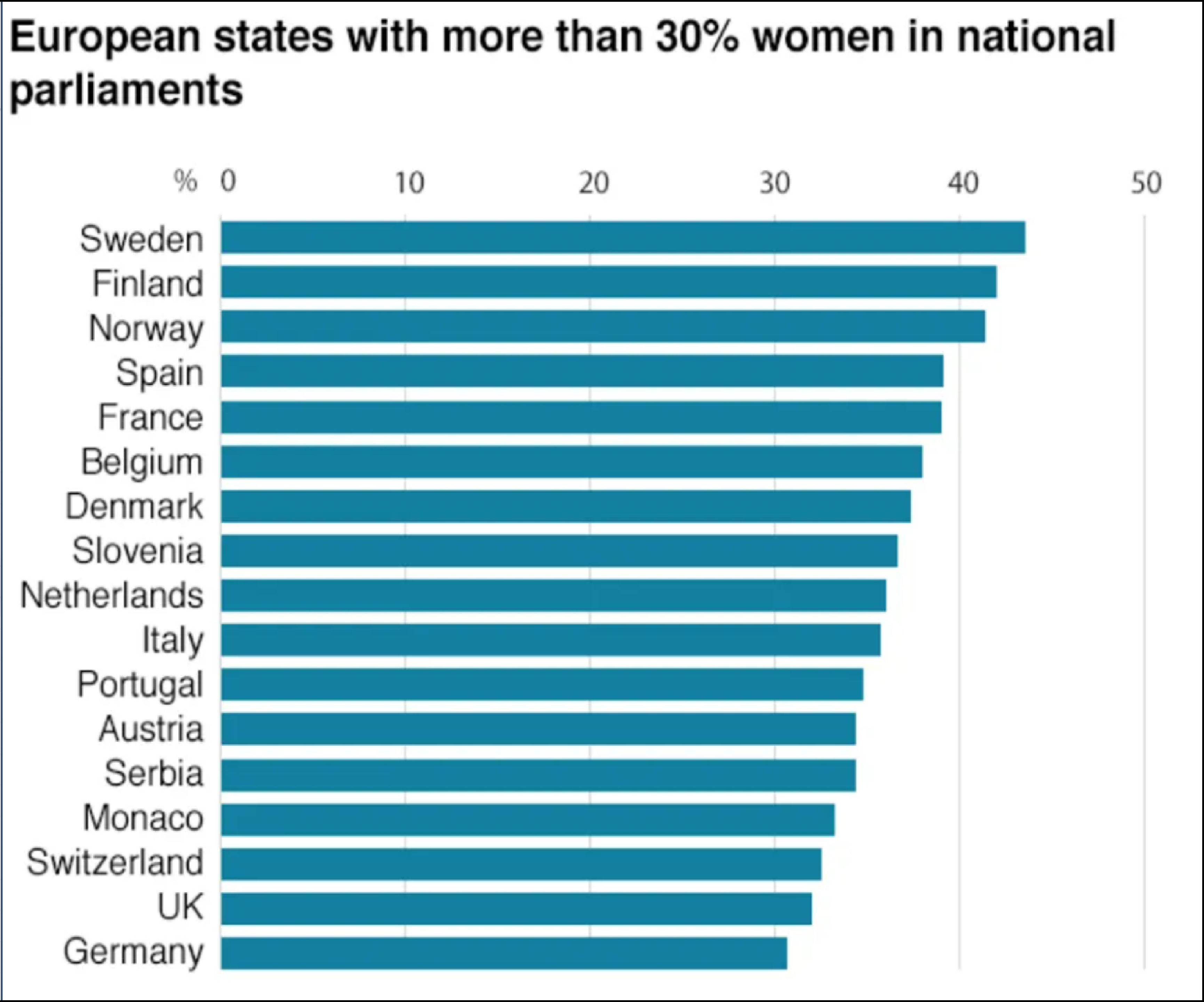
(Including turnover, except data for the 117th Congress are for the beginning of the Congress)



Source: Figure compiled by CRS, based on http://history.house.gov/Exhibition-and-Publications/WIC/Women-in-Congress/.



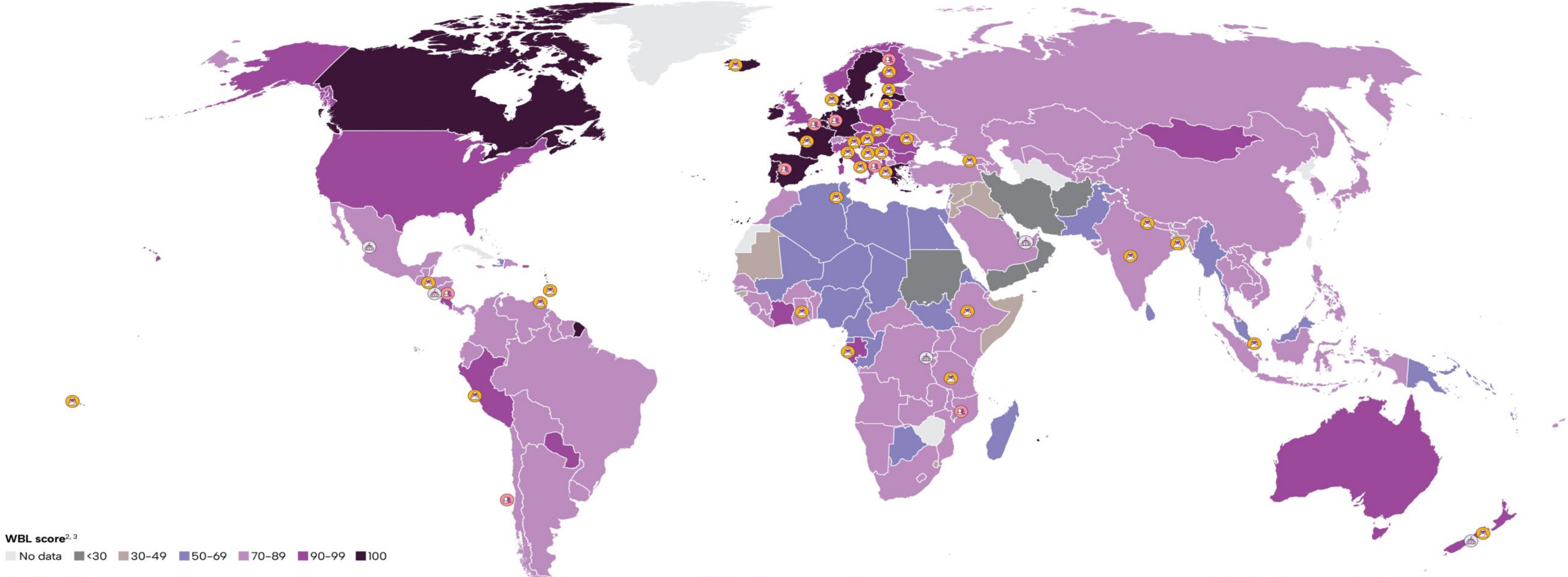
How Many Women Sit In Parliament?
Rwanda, Bolivia, and Cuba have the world's highest share of women in parliament. How do other countries compare?
Men in parliament
Women in parliament
Country
80%
40%
0
40%
80%
Rwanda 38.7% 61.3%
Bolivia 46.9% 53.1%
Cuba 51.1% 48.9%
Germany 63% 37%
Belarus 65.5% 34.5%
Serbia 65.6% 34.4%
Macedonia 68.3% 31.7%
U.K. 70% 30%
Afghanistan 72.3% 27.7%
Kazakhstan 72.9% 27.1%
Turkmenistan 74.2% 25.8%
Montenegro 76.5% 23.5%
Moldova 77.2% 22.8%
Bosnia-Herzegovina 78.6% 21.4%
Pakistan 79.4% 20.6%
Kyrgyzstan 80.8% 19.2%
USA 80.9% 19.1%
Tajikistan 81% 19%
Azerbaijan 83.2% 16.8%
Georgia 84% 16%
Uzbekistan 84% 16%
Russia 84.2% 15.8%
Ukraine 87.7% 12.3%
Armenia 90.1% 9.9%
Japan 90.7% 9.3%
Iran 94.1% 5.9%
World average of women in parliament is 23.4%



European states with more than 30% women in national parliaments
% 0 10 20 30 40 50
Sweden
Finland
Norway
Spain
France
Belgium
Denmark
Slovenia
Netherlands
Italy
Portugal
Austria
Serbia
Monaco
Switzerland
UK
Germany

- On Oct 21, 2025, Sanae Takaichi became the first female prime minister in Japan.


More women in leadership;
more women as national leaders.
AFP



31 countries have a woman as head of state[4, 5]

- Bangladesh
- Barbados
- Bosnia and Herzegovina
- Denmark
- Estonia
- Ethiopia
- Finland
- France
- Gabon
- Georgia
- Greece
- Honduras
- Hungary
- Iceland
- India
- Italy
- Lithuania
- Moldova
- Nepal
- New Zealand
- Peru
- Samoa
- San Marino
- Serbia
- Singapore
- Slovak Republic
- Slovenia
- Tanzania
- Togo
- Trinidad and Tobago
- Tunisia

But 154 countries have a man as head of state

5 countries have gender parity in parliament[6, 7]

- Mexico
- New Zealand
- Nicaragua
- Rwanda
- United Arab Emirates

But Yemen still has no women in parliament

14 countries have achieved legal equality of economic opportunity but 3 countries still have a WBL score below 30 points[3]

8 countries have 50% or more women as ministers[8]

- Albania
- Belgium
- Chile
- Finland
- Germany
- Mozambique
- Nicaragua
- Spain

But 7 countries have no women as ministers

- Azerbaijan
- Lebanon
- Papua New Guinea
- Saudi Arabia
- Sri Lanka
- Vanuatu
- Yemen

1 Includes only member states of the United Nations
2 2022
3 Total N: 185 countries
4 Head of State (excl. countries with monarchy-based system) and Head of Government
5 Total N: 193 countries
6 Total N: 186 countries
7 Seats in lower or single house
8 Total N: 182 countries

Curiosity: Female leaders and their impacts

- Impact in different domains
  - Fashion, Art, Education, Culture
  - Social Work, Community Services
  - Business & Management
  - Public Sectors & Politics
- How have the women leaders changed people's mind?
  - Specifically, how does their presence change the way we think about females and gender equality?
  - There are more women **national leaders** than ever – how do they matter?

Literature and Research Questions

Studies of Women Leadership &
Confidence in Women Leadership Capacity

Sociology

- Gender norm shift
- Labor participation and empowerment

Political science

- Electoral system
- Descriptive representation
- Public opinion change

Women Leadership Studies

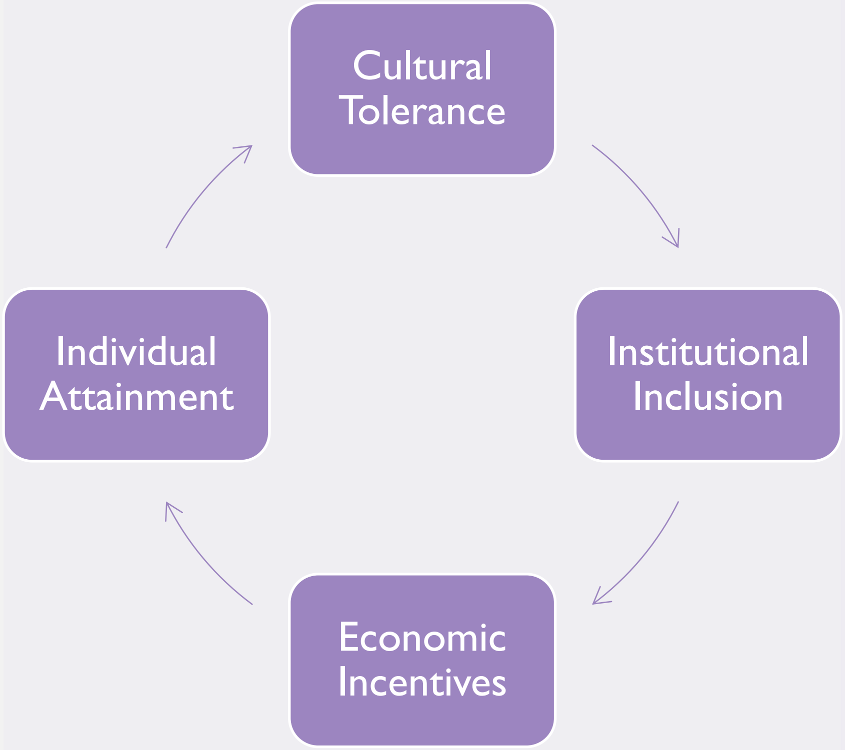
Management

- Gender and productivity
- ‘Glass ceiling’ & ‘Glass cliff’

Psychology

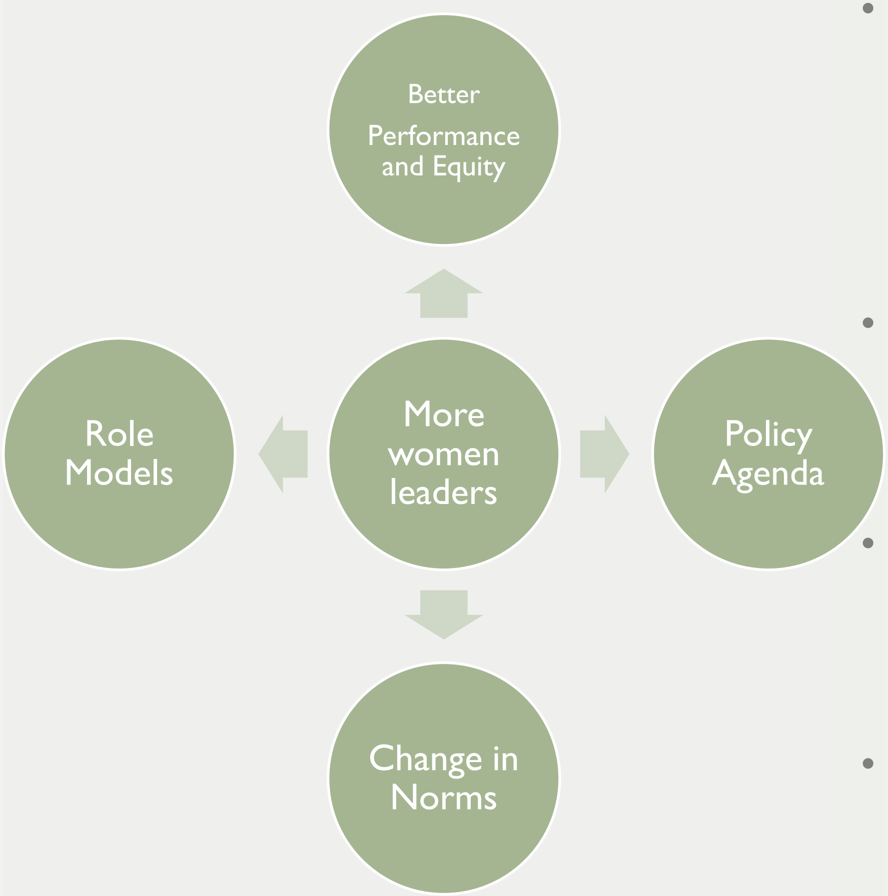
- Exposure effects
- Peer influences

Literature (1) Contributors to Women's Rise in Leadership Positions



- (1) Cultural and Societal Shifts (in Norms, Media, etc.)
  - Ridgeway(2011 ASR) argues that evolving societal attitudes reduce bias against female leaders
  - Paxton et al. (2007 JOP) indicates that increased female politicians inspire corporate leadership changes.
  - Eagly & Carli (2003 LQ) notes that positive portrayals of women leaders in media shift perceptions
- (2) Policy and Organizational Changes
  - Bertrand et al. (2019 AER) shows that legal frameworks (e.g., Title IX, board gender quotas) improve leadership diversity
  - Ely & Thomas (2001 AMJ) highlights that formal diversity programs help women advance.
  - Dobbin & Kalev(2016 ASQ) show that affirmative action policies in firms and governments increase female representation
  - Leslie et al. (2012 JAP) finds that work-life balance policies reduce high-potential women's struggle
- (3) Economic and Market Forces
  - Dezsö & Ross(2012 SMJ) finds that firms with female leaders perform better, driving demand
  - Kramer et al. (2006) highlights shareholder activism pushing for gender-balanced leadership
- (4) Individual-Level Factors (Composition, Cohort Replacement)
  - Goldin (2014) links higher education attainment among women to leadership pipeline growth
  - Ibarra et al. (2010 OrgSci) finds that access to influential networks accelerates women's advancement

Literature (2) Consequences of Women's Rise in Leadership Positions



- Economic & Firm Performance
  - Better governance: Female board members reduce excessive risk-taking (Review of Financial Studies, Adams & Ferreira, 2009).
  - Pay equity: Firms with women leaders have smaller gender pay gaps (AER, Bertrand & Hallock, 2001).
  - Innovation; Decision-Making; ESG agenda; etc.
- Political & Policy Shifts
  - Policy priorities: Female legislators prioritize health/education (APSR, Chattopadhyay & Duflo, 2004).
  - Corruption: Higher female political representation lowers corruption (JP, Dollar et al., 2001).
- Cultural and Societal Effects
  - **Cultural change**: Critical mass of women leaders reduces gender bias (AJS, Cohen et al., 1998).
  - **Role models**: Female leaders boost girls' career aspirations (Science, Beaman et al., 2012).
- Psychological Impacts
  - **Stereotype reduction**: Visible women leaders weaken bias (JPSP, Dasgupta & Asgari, 2004).
  - Backlash: Assertive women face likability penalties (Org Science, Rudman & Phelan, 2008).

Existing Works

- Previous works have focused on female business leaders and regular politicians and their impacts on people especially girls – overall, they do find a role model effect for young girls to see female leaders, though with various qualifications
  - Campbell, David E., and Christina Wolbrecht. "See Jane run: Women politicians as role models for adolescents." The Journal of Politics 68.2 (2006): 233-247.
  - Olsson, Maria, and Sarah E. Martiny. "Does exposure to counter-stereotypical role models influence girls' and women's gender stereotypes and career choices? A review of social psychological research." Frontiers in psychology 9 (2018): 392862.
  - Arvate, P. R., Galilea, G. W., & Todescat, I. (2018). The queen bee: A myth? The effect of top-level female leadership on subordinate females. The Leadership Quarterly, 29(5), 533-548.
  - Bos, Angela L., et al. "This one's for the boys: How gendered political socialization limits girls' political ambition and interest." American Political Science Review 116.2 (2022): 484-501.

Literature (3) Value consequences: Women Leadership and Public Opinion towards Women in general

- Overall, since the past few decades, there has been voices demanding and practices pushing for more women in leadership roles.
  - Rationale: More women leaders → more opportunities to be exposed to women leadership -> gender equality; representation; role model; empowering…
- Research Questions:
  - Does exposure really lead to higher confidence in women leadership capacity?
  - Are all kinds of exposure the same?
    - Symbolic appointment; "descriptive representation"

What about women as national leaders?

Would that differ from other settings?

Research Questions

- No prior research has focused on the role of female national leaders, especially head of states, on changing people's perception towards females and female leadership;
- Prior research has several unanswered questions:
  - (1) Women **Head of States** and Their Effects (Top Level Females)
  - (2) **Substantive vs. Symbolic** Leadership? (Which matters more?)
  - (3) Female **Empowerment** vs. Male **Enlightenment**? (Who changed more?)
  - (4) How would gender, duration of exposure, types of leaders **interact** in shaping public opinion towards females?

Research Hypotheses

RH 01: Duration Matters

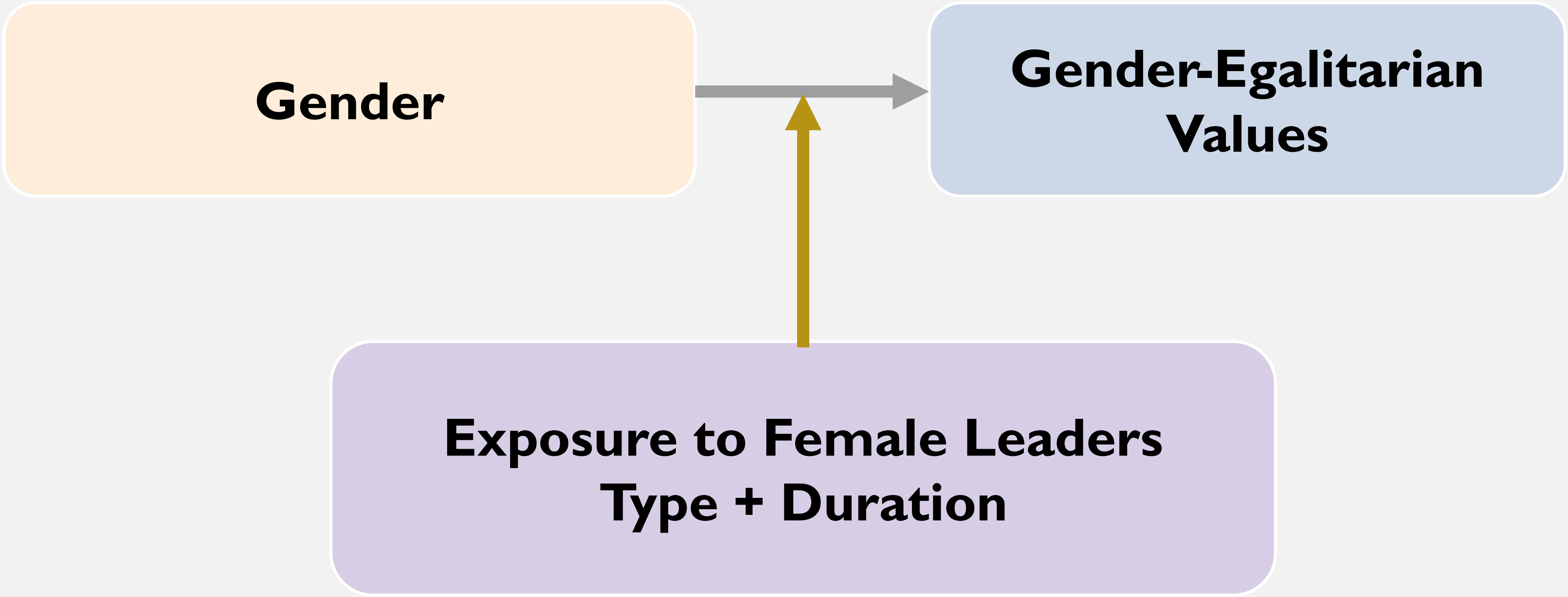
No Exposure → Short Exposure → Long Exposure

RH 02: Types of Leader Matters

Substantive Leader Matters More

Symbolic Leader Matters Less

RH 03: Gender and Exposure Effects



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Research Hypotheses (RH)

- Duration
  - (1) Longer exposure is associated with more gender-equal values
- Type
  - (2) Substantive female leaders (实质型元首) are effective in encouraging gender-equal values than ceremonial/symbolic leaders (名义型元首)
- Gender
  - (3) Females are overall more gender-equal, yet males changed more under exposure to female leaders

Data and Methods

Data: Women Head of States (WHOS)

- Types: Substantive WHOS vs. Symbolic (Ceremonial) WHOS




- Duration: Short vs. Long

50 days

16 years

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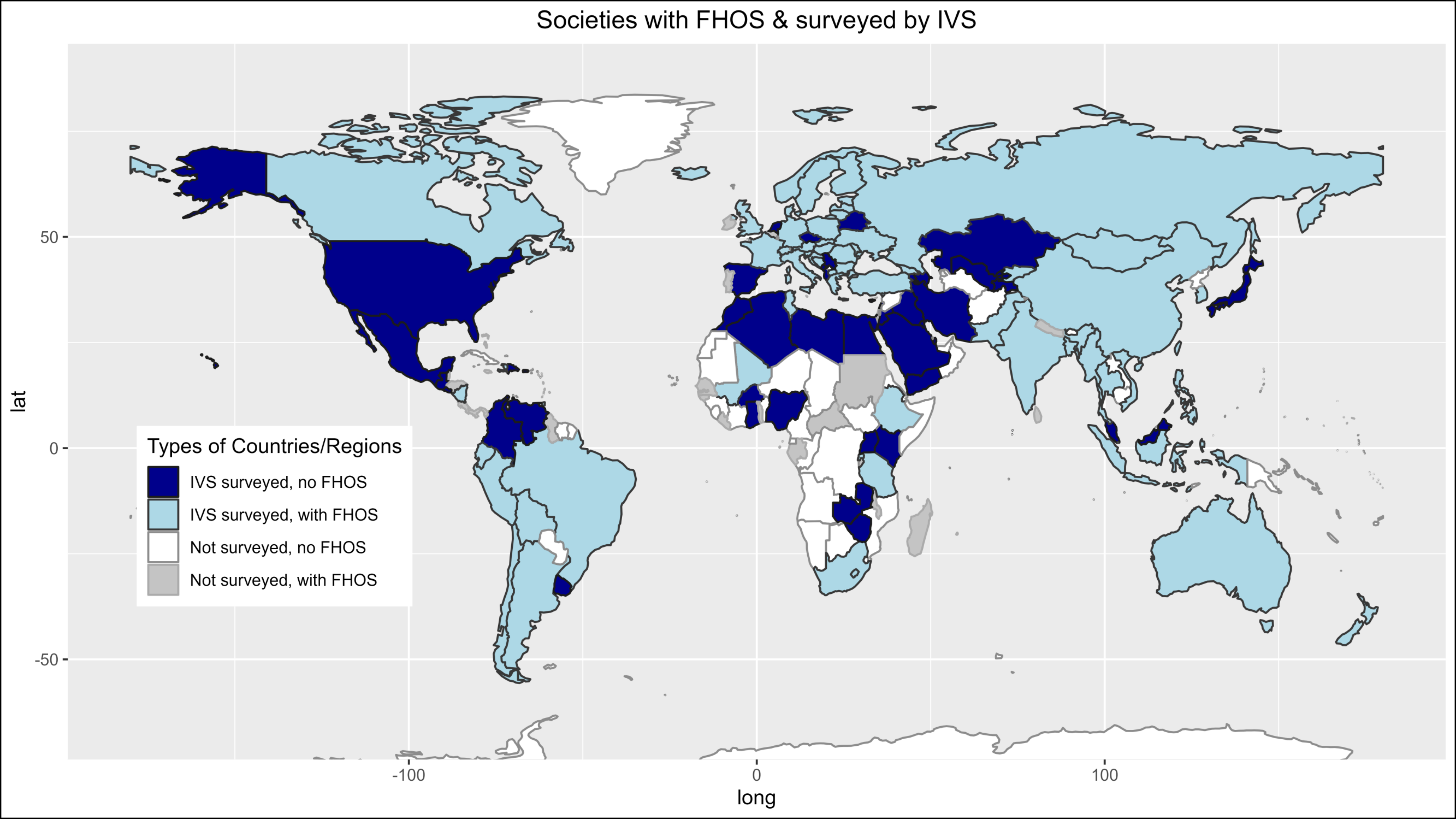
Data: Women Head of States (WHOS)

- We collected all WHOS profiles (459 individuals from 90+ countries, regions, and territories), ranging from January 1940 to 2023.

- We then merge the WHOS (1940-2023) data with the World Values Survey data (Wave 3-7, 1995-2022).

- Within the overlapped temporal and geographical scope, there are 186 female national leaders coming from 60 countries, regions, and territories.


Societies with FHOS & surveyed by IVS
Types of Countries/Regions
IVS surveyed, no FHOS
IVS surveyed, with FHOS
Not surveyed, no FHOS
Not surveyed, with FHOS
lat
long
50
0
-50
-100
0
100

Data

- Individual Level (Integrated Values Survey = WVS + EVS)
  - Seven Waves

Years	1981-1984	1990-1994	1995-1998	1999-2004	2005-2009	2010-2014	2017-2022
WVS Waves	WVS1	WVS2	WVS3	WVS4	WVS5	WVS6	WVS7
EVS Waves	EVS1	EVS2		EVS3	EVS4		EVS5

  - More than 412,890 obs. from 306 surveys and 110 countries/regions
- Aggregate Level:
  - Economic vars: World Bank; IMF; UNDP; Solt (2016)
  - Political vars: Freedom House; Polity V data; V-Dem project;
  - Cultural and other vars: Shalom Schwartz (2001); Samuel Huntington (1993)

Data: Outcome Variable and Items

- Four Items; Cronbach's alpha > 0.83 (dv1) & > 0.77 (dv2)

Value Items (Range: 1-4) 4 = strongly disagree (support gender equality) 1 = strongly agree (favoring male dominance)	Mean	SD	Min	Max
Item 1: Men are better political leader	2.64	0.97	1	4
Item 2: Men are better business leader	2.54	0.99	1	4
Item 3: Hire men first when jobs are scarce	2.63	1.36	1	4
Item 4: Educate boys first when edu opportunity are scarce	3.02	0.91	1	4
DV1: (items 1&2) Conf in women leadership	2.60	0.87	1	4
DV2: (items 1 - 4) Gender equality	2.71	0.81	1	4

confidence in women leadership

support for gender equality

Data: Predictors

- Age, Gender, Marital Status, Educational Levels, Income Levels
- Exposure to Women Head of States
  - YES or NO (Dummy)
  - Duration of Exposure (Continuous, Months / Terms)
  - Types of WHOS (Substantive vs. Symbolic)

Descriptive Statistics and Correlation Matrix

Table 1. Means, SDs and the Correlation Matrix of Variables

Variables	Mean	SD	1	2	3	4	5	6	7	8
1 Gender (Male=1)	0.48	0.50	1.00							
2 Age in years (18-99)	42.25	16.40	0.00	1.00						
3 Marital (Married=1)	0.75	0.43	-0.07***	0.48***	1.00					
4 Education (Univ. = 1)	0.19	0.39	0.02***	-0.03***	-0.03***	1.00				
5 Exposed to WHOS (Yes = 1)	0.37	0.48	-0.01***	0.13***	0.08***	-0.00	1.00			
6 GDP per capita ($1,000)	21.86	18.16	-0.02***	0.23***	0.02***	0.12***	0.12***	1.00		
7 Gender Inequality (0-1)	0.77	0.15	-0.02***	0.21***	0.02***	0.04***	0.19***	0.53***	1.00	
8 Liberal Democracy (0-1)	0.48	0.27	-0.01***	0.20***	0.01***	0.03***	0.15***	0.61***	0.76***	1.00
9 Confidence in Women Leadership Potential (1-4)	2.60	0.87	-0.15***	0.05***	-0.03***	0.09***	0.10***	0.34***	0.37***	0.35***

Note. $^{***}p < .001$; $^{**}p < .01$; $^{*}p < .05$. Gender: male = 1, female = 0. Marital status: married = 1, unmarried = 0. University educated: yes = 1, no = 0. to WHOS (yes = 1, no = 0) includes exposure to substantive and symbolic women head of state. Gender inequality and liberal democracy indices bot from 0-1, where higher values stand for more equal and liberal; confidence in women leadership ranges from 1-4 and higher values stand for more confidence.

Selection Problem: Propensity Score Matching (PSM)

- Countries and individuals that could elect WHOS systematically differ from countries that never elect any WHOS.
  - Countries: in terms of economy, political systems, culture, religion, social norms, population, ideology, …
  - Individuals: in terms of beliefs, gender values, educational levels, lifestyles, critical life choices, occupation, income, social statuses, …

- PSM – nearest neighbor algorithms
  - With R package '**MatchIt**" by Gary King and colleagues at Harvard
  - Ho, D., Imai, K., King, G., & Stuart, E. A. (2011). MatchIt: nonparametric preprocessing for parametric causal inference. Journal of statistical software, 42, 1-28..
  - To match cases that are similar on the following variables: GDP pc, Political Freedom, Age, Gender, Marital Status, Socio-economic Status, Education.

Instrumental Variable (IV) & 2SLS design

- Instrumental Variable
  - Time-Lagged IV
  - Sex Ratio at Birth, 20 years ago (SRAB $_{t-20}$)
  - Data from: World Bank data on SRAB during 1960 - 2025

- Two-Stage least squares (2SLS) regression
  - With R package '**ivreg**" by John Fox et al. (2020)
  - Fox, J., Kleiber, C., & Zeileis, A. (2020). Diagnostics for 2SLS regression.

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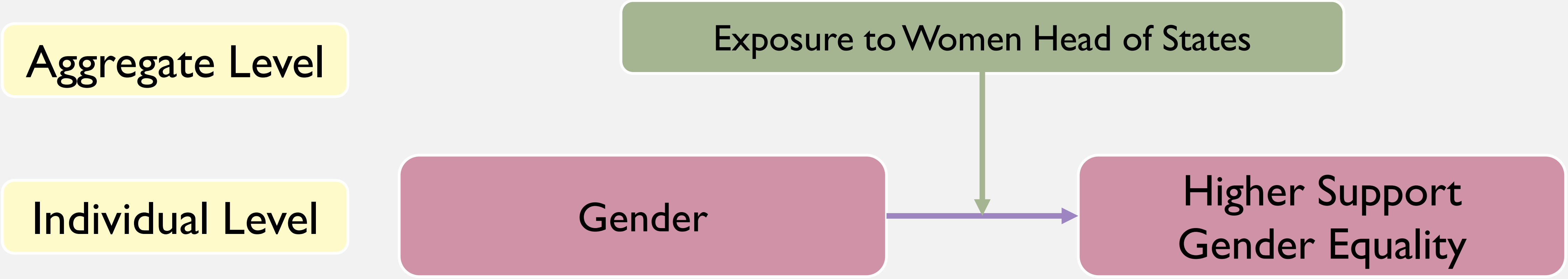
Regression Models

- Fixed-Effects Models (with country FE, year FE, focal predictors)

- Interaction Models: type of exposure * gender with FE



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Sensitivity and Robustness

- Women Head of States (WHOS) observations
  - Duration of WHOS: at least 30 days (7, 14, 90, 180, 365)
  - Inclusion/exclusion of various types of WHOS (vice WHOS/non-elected/appointed WHOS)

- Aggregate-level cases
  - Exclusion of possible outliers (extremely rich/poor countries)
  - Lagging variable: different ($t - n$)
  - Different control variable alternatives (political; gender inequality)

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Findings

Fixed Effects Model (Unmatched + PSM matched)

Table 2. Models Estimating Perceptions in Women's Leadership Potential with Types of Exposure to WHOS in Study 1

	Model 1	Model 2	Model 3	Model 4
Country & year FE	*Yes*	*Yes*	*Yes*	*Yes*
Control variables	*No*	*Yes*	*No*	*Yes*
PSM matched sample	*No*	*No*	*Yes*	*Yes*
Intercept	-24.58***	-15.80***	-24.39***	-14.94***
	(0.37)	(0.59)	(0.44)	(0.73)
Individual-Level Controls:				
Age in years (18-99)		-0.00***		-0.00***
		(0.00)		(0.00)
Male (Female = 0)		-0.26***		-0.24***
		(0.00)		(0.00)
Married (Not Married = 0)		-0.03***		-0.03***
		(0.00)		(0.00)
University Degree (No = 0)		0.14***		0.14***
		(0.00)		(0.00)
Aggregate-Level Controls:				
GDP (in $1,000 USD)		0.01***		0.01***
		(0.00)		(0.00)
Gender Equality Index (0-1)		-0.18***		-0.48***
		(0.04)		(0.05)

Table 2. Models Estimating Perceptions in Women's Leadership Potential with Types of Exposure to WHOS in Study 1

	Model 1	Model 2	Model 3	Model 4
Liberal Democracy Index (0-1)		0.24***		0.33***
		(0.02)		(0.03)
Focal variables	*Displayed with No WHOS = 0*			
symbolic WHOS	-0.06***	-0.08***	-0.01	-0.07***
	(0.01)	(0.01)	(0.01)	(0.01)
Substantive WHOS	0.04***	0.05***	0.08***	0.05***
	(0.01)	(0.01)	(0.01)	(0.01)
	Displayed with symbolic WHOS = 0			
No WHOS	0.06***	0.08***	0.01	0.07***
	(0.01)	(0.01)	(0.01)	(0.01)
Substantive WHOS	0.11***	0.13***	0.09***	0.12***
	(0.01)	(0.01)	(0.01)	(0.01)
R^2	0.25	0.28	0.24	0.27
Adj. R^2	0.25	0.28	0.24	0.27
Num. of observations	379815	379815	280792	280792

Note. Numbers in brackets are standard errors. Symbolic WHOS group was used as the reference group. The sample size of Model 4 differs from previous samples due to matching procedures. $^{***}p < .001$; $^{**}p < .01$; $^{*}p < .05$.

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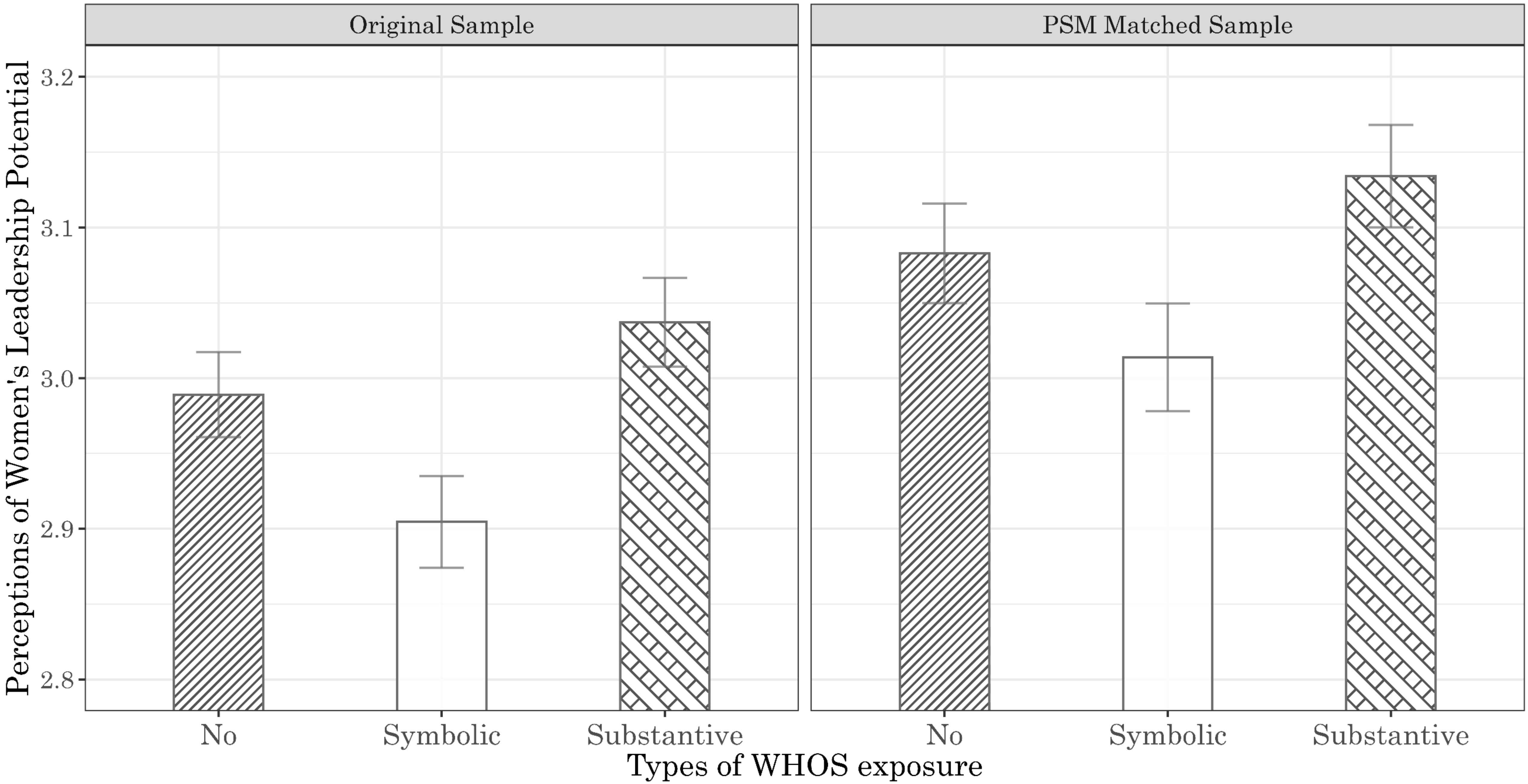
Original Sample
PSM Matched Sample
Perceptions of Women's Leadership Potential
3.2
3.1
3.0
2.9
2.8
No
Symbolic
Substantive
No
Symbolic
Substantive
Types of WHOS exposure

Table 3. The Influence of WHOS exposure (No vs. Yes & Symbolic vs. Substantive) on Perceptions in Women's Leadership Potential: Results of the Instrumental Variable Analysis.

Dependent variable	Model 5		Model 6	
Predictors	Exposure (first stage)	Perception (second stage)	Exposure (first stage)	Perception (second stage)
Sex Ratio at Birth $_{t-20}$ (100~118)	-0.025***		-0.032***	
	(0.000)		(0.000)	
Exposed to WHOS (No=0)		0.664***		
		(0.064)		
Exposed to Substantive WHOS (Symbolic WHOS=0)				1.207***
				(0.087)
Control variables	Yes	Yes	Yes	Yes
Country FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Num. of Observations		280792		140396
Weak Instrument Test	3376.8***		5077.3***	
Wu-Hausman Test	105.1**		194.8***	

Notes: ***$p < .001$; **$p < .01$; *$p < .05$. Both models are fitted on PSM matched sample. Model

Estimates from IV and 2SLS analysis

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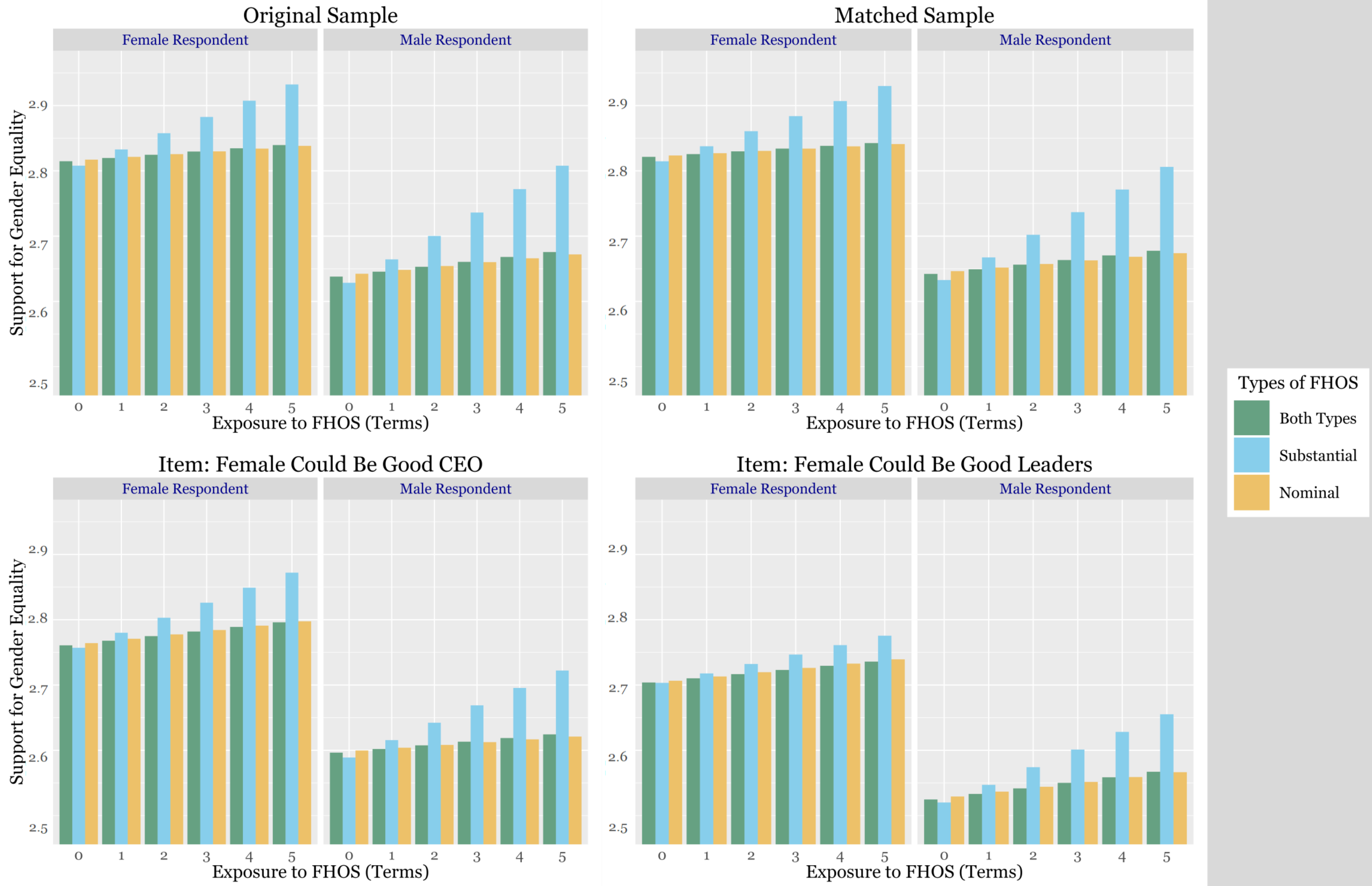

Original Sample
Matched Sample
Item: Female Could Be Good CEO
Item: Female Could Be Good Leaders
Female Respondent
Male Respondent
Support for Gender Equality
Exposure to FHOS (Terms)
2.5
2.6
2.7
2.8
2.9
0
1
2
3
4
5
Types of FHOS
Both Types
Substantial
Nominal

Summary

Summary of Results

- Exposure to female national leaders would increase the overall support of gender equality and confidence in female leadership
- Such exposure effects are stronger when the exposure is longer (duration matters)
- Such effects are stronger among males than females, supporting the enlightening thesis over the empowerment thesis
- Such exposure effects are stronger when "real" leaders in position; symbolic WHOS associate with less confidence in women

Contributions and Limitations

- Contributions
  - Women leaders -> women political leaders -> **women national leaders**
  - The distinction of two types -> **symbolic vs. substantive leaders**
- Limitations
  - Selection bias still exists
  - IV – still not ideal – may violate the ‘exclusion restriction’ criterion

Policy Implications

- We need more women leaders;

- If we want to maximize their positive effects on people's overall gender egalitarian disposition, we need to have them as **substantive** leaders, rather than **symbolic** ones.

妇女能顶半边天

Women hold up half the sky
-- Chairman Mao

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Future Agenda

- Experiment / Survey Experiment;

- Life course perspective: exposure during different stages (e.g., 3-12; or teenage years13-19, or young adulthood)

- Female Leaders elected/appointed during crisis moments and consequences ('glass cliff' literature)

Take-Away Messages

- Exposure to female national leader matters
  - Female in real and powerful positions matters (**type**)
  - The longer they serve, the more impactful (**duration**)
  - Perceptions can change – especially for guys (**gender / male enlightenment effect**)

Thank You!

Questions and comments welcomed

Email: huiquanzhang@um.edu.mo

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